AI project Group 05
The concerns are
1. Hyper personalied Marketing
Things to do to solve
• Collect customer data - purchase history, browsing history, preferences, demographics etc
• AI/ML on customer data to segregate customer groups & individual preferences
• Recommend related product & offers to each customer.
• Personalize Emails, Ads, notifications & promotions.
• Constantly analyse and improve your campaigns.

Goal: Right product - Right customer - Right time.

2. Analysing sentiments of customer feedback
Things to do to solve
• collect all reviews, survey, social media comments, complaints etc
• Use NLP for feedback analysis
• classify the feedback as positive, negative or neutral
• identify common problems & customer expectation.
• Provide dashboard / reports to marketing & management team.

Goal: To know what customer likes, dislikes & expectations.

3. Improve customer service
Things to do to solve
• Implement AI based chatbot to provide 24*7 basic support.
• Use customer data for personalized response.
• Automate FAQs, query of order/status, common complaints.
• Escalate complex issues to human employees.
• Analyze customer service conversation to identify common issues.

Goal: Faster responses with personalisied support & more focus on customer satisfaction.

4. Data driven decision making
Things to do to solve
• collect & combine data of customer, sales, marketing data etc.
• Using AI to identify trends and patterns in customer behavior
• Leveraging predictive analytics to anticipate customer needs and demand
• Creating AI-driven dashboards and reports
• Leveraging insights to optimize product, pricing, and marketing strategies.

Goal: To enable faster and more informed business decisions based on customer data.

5. Continuous improvement
Things to resolve:
• Continuously monitoring of customer response, campaign performance
• Monitor KPI's (Engagement, Conversion, Customer satisfaction etc.)
• Leverage AI to identify what's working and what needs improvement.
• Experiment with different marketing strategies and their outcomes.

• Continuously update AI models with new customer data.

Goal: Continuous improvement of marketing performance and customer satisfaction.

1. Hyper personalised marketing

AI Architecture:

Implementing Artificial Narrow Intelligence (ANI) with limited memory systems, which learn from past actions of users and purchase patterns, constantly.

Machine learning and Deep Learning:

Clustering & RFM Segmentation

Grouping customers by LTV (Lifetime value), Price Elasticity and Product Categories, (Unsupervised Learning)

Optimization and Decision Making

A Search & Heuristic Search

Dynamic calculation of the shortest (or low cost) delivery path for 'on demand' (ex. Swiggy like) deliveries and logistics routing.

Simultaneous move game theory

Modeling competition discount pricing (ex: mimic competitor flash sales. Eg. Flipkart /Amazon)

Collect all types of customer data (structured, unstructured, semi-structured). All three primary types of data (Structured: Demographic, CRM Tables. Unstructured: social media history, text of email. Semi-structured: JSON website clickstream).

Finds customer segments, individual preferences: Unsupervised learning (Clustering) to extract hidden structures from unlabelled data and cluster customers into specific groups based on common buying behaviour/habits.

Finds relevancy products and offers to recommend. Basically uses knowledge graphs to represent semantic relationships between entities (ex: customer product category has brand), which can be used to power recommendation engine (as done by Flipkart, Amazon). Uses Market basket analysis (Unsupervised learning) to find what items are often bought together.

Hyper personalized emails, ads, promotions: Generates hyper-personalized marketing copy, product descriptions, email etc. automatically using generative AI (LLM) - at scale for each individual customer's context.

Continuously checking results and improves campaigns: Uses reinforcement learning to learn from continuous user feedback (click, purchase). Table 1

2. Sentiment Analysis from customer feedback.

Categorizes user feedback into specific attributes (ex. Quality of packaging, fragrance, price, skin sensitivity etc.) instead of generic ratings.

Reasoning models:

Inductive reasoning - to derive predictive insights from past samples of reviews

Abductive reasoning -

Diagnosing reason for sudden dip in product repeated purchases, by relating feedback to potential operations issues.

Fuzzy Logic to Handle Real World Ambiguity
No binary (satisfied v/s unsatisfied) sentiment but Fuzzy Membership Functions (e.g., scale 0.0 to 1.0 e.g. 0.75 moderately satisfied) to properly depict subtle qualitative feedback.

Gather Reviews/surveys/ social media comments – Process large volumes of Unstructured Data (80 - 90% of all global data) which has no fixed schema.

Classify (positive/negative/neutral) – Supervised Learning (classification). The AI learns to map (this is where the action happens) by learning from historical human-labeled examples of positive and negative reviews. By doing so, it can automatically categorise new unknown texts.
Identify common problems and customer expectatons - Unsupervised Learning (Topic modelling) – scanning thousands of company's documents & reviews (unlabelled) to organically identify key themes and common issues. Table 1

3. Enhanced Customer Service
Tier 1 – (Automated bot)
Implement rule-based/intent detection triage bots for common requests (order status, return policies etc.)
Tier 2 (GenAI & limited memory assistants)
Fine-tuned (LLMs) - store customer history for sessions, to provide empathetic & contextual solutions without repeating yourself manually.

Escalation path (Graph search)
Provide a Breadth-First-Search (BFS) in internal escalation tree for departments to get the right expert for a complex complaint through the shortest path of escalation.
Early stage sentiment tuning (Theory of mind concepts)
Detect frustration and change in tone during interactions. Automatically escalating urgent issues to high ranking human agent.

AI chatbot for 24/7 basic support
Rule based systems. For simple, high-reliability, IF-THEN -based rigid logic chatbot for simple FAQs resolution.
Generative AI. For more flexible natural language understanding.

Provide personalized response using customer data
Semantic technologies and Ontologies to understand the deep semantics and context of the customer query instead of using brittle keyword search.

Automated FAQs, order/status, or common complaints
Upgrade from reactive generative chatbots to Agentic AI.
Agentic AI instead of just respond to text prompt, it uses external tools/APIs to autonomously execute multi-step tasks. E.g. Look up an order. Check delivery status. Process refund for customer.

Escalate complex issue to human employees
Responsible AI: Principle of human oversight. Human in control of high-stakes and nuanced or very emotional customer service escalations. AI does not have true empathy (Theory of mind) Table 1

4. Data-Driven Decision Making:
- Aggregation of customer, sales and marketing data
- Data Type and representations
- We have to be able to aggregate all three main data types:
- Structured Data (SQL table sales data, rows in CRM)

Knowledge Graphs and Ontologies: This is the make all of this data as one. Real-time customer 360 data platform. Triples (e.g. customer A - Bought - Product B) data structures allow the system to create a semantic network that provides deep and contextual understanding of the relationships between disparate data bases for the AI to comprehend.
Use AI to discover trends and patterns in customer behaviours
Unsupervised Learning: Used to discover hidden structure in unlabelled customer data, without human intervention.
Clustering: K-Means type of algorithms used to group customer in segments based on similar buying behaviour/habits
Market Basket: Uses association rules and identify affinity patterns. Identify which products are often bought together to uncover shopping trends.
Use predictive analytics to forecast customer needs and demand
Supervised Learning: Learning the mathematical mapping from past inputs given to desired outputs.
Regression Trees: Predicts continuous numerical targets. Example Predicts exact sales volumes. Or next quarter demand sensing.
Classification: Predicts discrete categories. Example predicts risk of churn. High/Low risk for a specific segment of customer
AI Demand Sensing Platform: A specialized tool that uses internal historical data combined with external signals to predict real time demand.

Creating AI-powered dash-boards and reports
1. Explainable AI (XAI) & Decision Trees: Business Dashboards need transparency. Decision Trees as a "white-box" model to deliver XAI Reporting Layer where managers can see exactly the IF/THEN logical splits that resulted in a given forecast or insight.
2. Gen AI for Analytics: Enabling non-technical execs to simply query complex datasets in plain English and automatically generate narrative summary of the data.
Use insights to improve products, pricing, marketing etc.
3. Game Theory: For optimal pricing in competitive markets. Example Use of "Simultaneous move games" (e.g. Prisoner's dilemma). Enable AI to predict competitors response to pricing changes and hence enable optimal dynamic pricing strategies to be implemented without sparking destructive price wars. Table 1

5. Continuous Improvement:
Continuously monitor customer responses/campaign performance.
Agentic AI. Closed loop architecture. Works as an autonomous agent in continuous loop. Perceive (monitor for click/engage on campaign) Reason (Evaluate success) Act (Adjust spend) Reflect (Measure new outcome) Probabilistic Reasoning (Bayes' Theorem). The AI will constantly update the Probability of success of your campaign based on the feedback from new customers. So it can tune the posterior probabilities for the live data.
Track your KPIs (e.g. engagement, conversion, customer satisfaction)

Fuzzy Logic: Conversion is a clear indicator. But customer satisfaction may be a fuzzy term. Fuzzy logic is about dealing with partial truths. So the AI can score how satisfied your customer is with a degree of membership (e.g. 0.85 mostly happy), instead of a strict true/false dichotomous class.
Use AI to see what's good and what's bad
Abductive Reasoning: The AI, when seeing an unexpected KPI value (e.g. "Sales dropped by 40%"), will use abductive reasoning to work out the most likely root-cause/best explanation (e.g. "A competitor came out with similar product at lower price").
Decision Tree Feature Importance. You can see all the important features (e.g. email subject line, amount of discount) that influence conversion the most, based on the analysis of branches of the decision tree.
Try various marketing strategies and compare.
Reinforcement Learning (RL): The solution for testing. Use the Multi-Armed Bandit concept - balance explore (try out new and untested marketing ad variants) with exploit (push the budget to the highest performing ad so that more immediate conversions happen) throughout the AI.

Automated A/B Testing & Performance Optimiser: Dynamically shift traffic in real time to better performing creatives based on RL feedback signals.

Continually update AI models with new customer data.

Continuous Model monitoring and drift detection system - to prevent AI from having "Metrics Mismatch" / predictive decay - by automatically tracking performance and forcing re-training in response to shifts in customer behaviour. Table 1

Strategic Pillars & Value Stream (Top Flow)
The middle workflow. Bridging the tech roadmap into business impact tiers:
1. Hyper-Personalised Marketing (High ROI Impact): Use of K-Means clustering + Knowledge Graph triples + Multi-Armed Bandits to enable automated dynamic ad serving to boost top of funnel conversions.
2. Sentiment Analysis (Medium ROI impact). It involves using supervised NLP, topic modeling and fuzzy logic to track qualitative feedback, catch root cause product defects early, and protect customer brand equity.
3. Better Customer Service (High ROI impact). It involves using rule based triage bots, fine-tuned LLMs, and agentic task execution to deflect routine tickets, leaving only complicated escalations for human staff.
4. Data Driven Decision Making (High ROI impact). It involves using regression trees, demand sensing and game theoretic pricing models to forecast inventory requirements, avoid destructive price wars.
5. Continuous Improvement (Cumulative ROI impact). It involves using closed loop telemetry, Bayesian updates, Multi-Armed Bandit experimentation to keep compounding the returns and avoid model degradation over time. Table 3

Reference:-

https://www.ibm.com/think/topics/hyper-personalization
https://aws.amazon.com/what-is/sentiment-analysis
https://www.forbes.com/sites/kathleenwalch/2024/07/15/how-generative-ai-is-driving-hyperpersonalization
https://chattermill.com/blog/ai-sentiment-analysis-tools
https://gemini.google.com
https://aistudio.google.com
https://drive.google.com/drive/folders/1GWXMvKkGgExHUF3MH6ZVmKdOgHV8puHe?usp=drive_link

Appendix:-

Pillar / Objective	Primary AI / Algorithmic Technique	Data Input Type	Target Business Outcome
1. Hyper-Personalization	K-Means, Knowledge Graph Triples, Multi-Armed Bandit	Structured (SQL), Semi-Structured (JSON)	Higher conversion rate, dynamic ad serving
2. Sentiment Analysis	Supervised NLP Classifiers, Topic Modeling, Fuzzy Logic	Unstructured (Reviews, Social feeds)	Granular attribute tracking, early defect detection
3. Customer Support	Rule-Based Trees, Fine-Tuned LLMs, Graph Search (BFS)	Unstructured (Chat text), CRM sessions	First-contact resolution, reduced ticket escalations
4. Decision Making	Regression Trees, Game Theory (Payoff Matrix), Heuristics	Multi-source (Sales, Web, Competitors)	Demand forecasting, price stability without margin wars
5. Continuous Learning	Agentic Loops, Bayesian Updating, Concept Drift Audits	Real-time campaign telemetry	Prevention of model decay, self-tuning budgets

Table 1

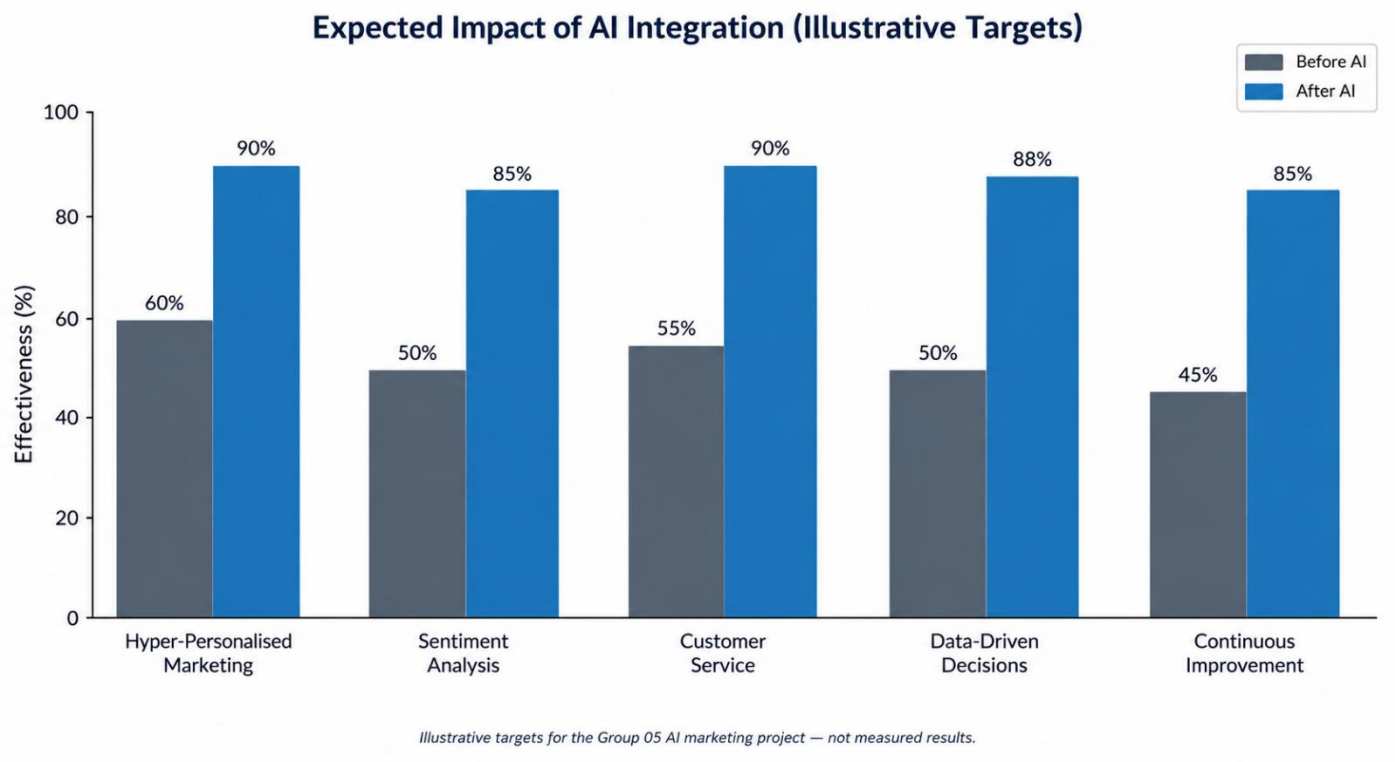


Table 2

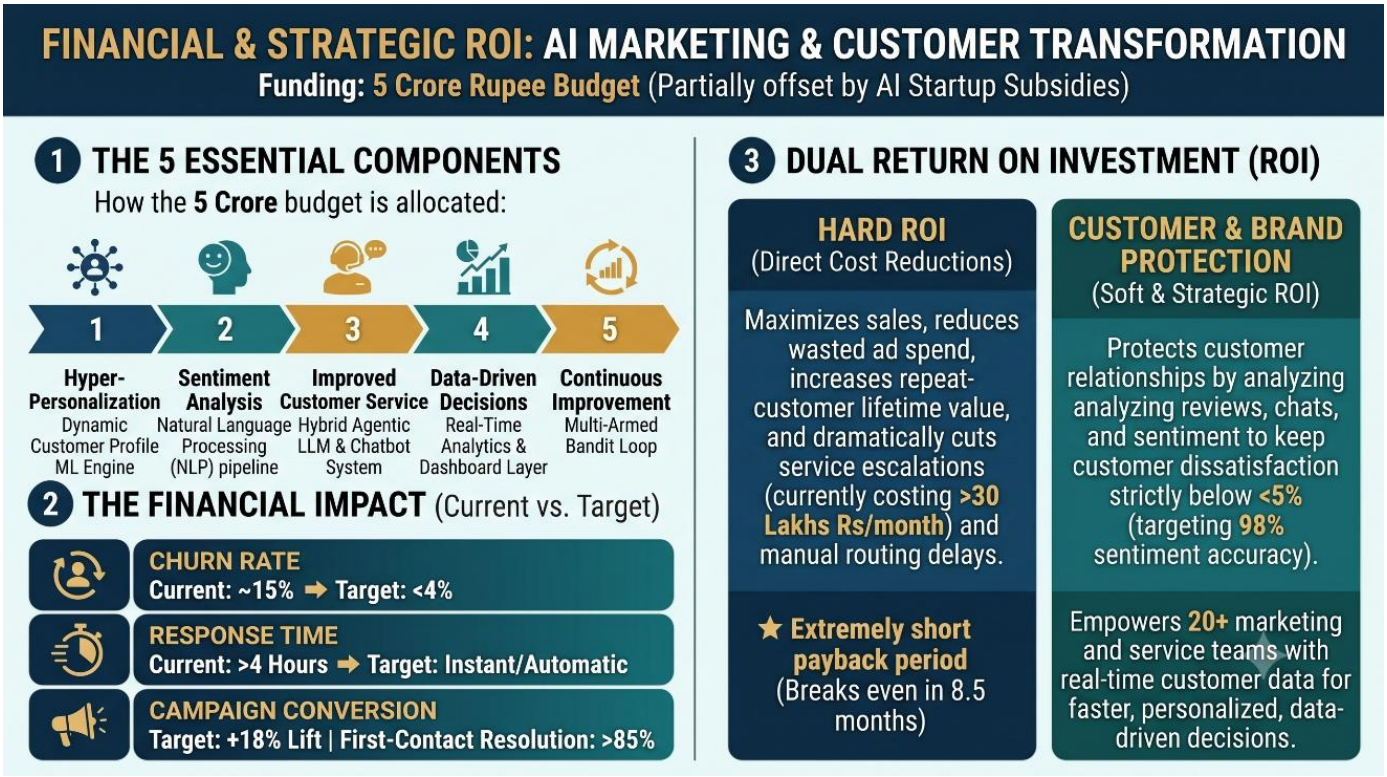

FINANCIAL & STRATEGIC ROI: AI MARKETING & CUSTOMER TRANSFORMATION
Funding: 5 Crore Rupee Budget (Partially offset by AI Startup Subsidies)
1 THE 5 ESSENTIAL COMPONENTS
How the 5 Crore budget is allocated:
1
Hyper-Personalization
Dynamic Customer Profile ML Engine
2
Sentiment Analysis
Natural Language Processing (NLP) pipeline
3
Improved Customer Service
Hybrid Agentic LLM & Chatbot System
4
Data-Driven Decisions
Real-Time Analytics & Dashboard Layer
5
Continuous Improvement
Multi-Armed Bandit Loop
2 THE FINANCIAL IMPACT (Current vs. Target)
CHURN RATE
Current: ~15% ➡ Target: <4%
RESPONSE TIME
Current: >4 Hours ➡ Target: Instant/Automatic
CAMPAIGN CONVERSION
Target: +18% Lift | First-Contact Resolution: >85%
3 DUAL RETURN ON INVESTMENT (ROI)
HARD ROI
(Direct Cost Reductions)
Maximizes sales, reduces wasted ad spend, increases repeat-customer lifetime value, and dramatically cuts service escalations (currently costing >30 Lakhs Rs/month) and manual routing delays.
★ Extremely short payback period
(Breaks even in 8.5 months)
CUSTOMER & BRAND PROTECTION
(Soft & Strategic ROI)
Protects customer relationships by analyzing analyzing reviews, chats, and sentiment to keep customer dissatisfaction strictly below <5% (targeting 98% sentiment accuracy).
Empowers 20+ marketing and service teams with real-time customer data for faster, personalized, data-driven decisions.

Table 3